

5.5 For each of the following two time-series regression models, and assuming MRI-MR6 hold, find $\text{var}(b_1|x)$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_1|x) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $x = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$.
 $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2(0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $x = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^T = \frac{r(1 - r^T)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

a.

$$s_{tt} = \sum t^2 - \frac{(\sum t)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2}{4} = \frac{T^3 - 1}{12}$$

$$\text{Var}(b_1) = \frac{\sigma^2}{s_{tt}} = \frac{12 \cdot \sigma^2}{T^3 - 1}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_1) = \lim_{T \rightarrow \infty} \frac{12 \cdot \sigma^2}{T^3 - 1} = 0$$

b.

$$s_{tt} = \sum (0.5^t - \bar{x})^2 = \sum 0.5^{2t} - T \bar{x}^2 = \sum 0.5^{2t} - \frac{(\sum 0.5^t)^2}{T} = \frac{0.25(1 - 0.25^T)}{1 - 0.25} - \frac{1}{T} \left(\frac{0.5(1 - 0.5^T)}{1 - 0.5} \right)^2 = \frac{1}{3}(1 - 0.25^T) - \frac{1}{T}(1 - 0.5^T)^2$$

$$\text{Var}(b_1) = \frac{\sigma^2}{\frac{1}{3}(1 - 0.25^T) - \frac{1}{T}(1 - 0.5^T)^2}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_1) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{\frac{1}{3}(1 - 0.25^T) - \frac{1}{T}(1 - 0.5^T)^2} = 3\sigma^2 \neq 0$$

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

a.

$$H_0: \beta_2 = 0$$

$$H_a: \beta_2 \neq 0$$

$$t = \frac{b_2}{\text{SE}(b_2)} \sim t(b_0)$$

$$t_0 = \frac{3}{2} = 1.5, \quad t_{0.025}(b_0) = 2$$

$$RR = \{|t| > 2\}$$

$$\because t_0 = 1.5 < 2 \quad \therefore \text{Do not reject } H_0: \beta_2 = 0$$

b.

$$H_0: \beta_1 + 2\beta_2 = 5$$

$$H_a: \beta_1 + 2\beta_2 \neq 5$$

$$\text{SE}(b_1 + 2b_2) = \sqrt{3 + 2^2 \cdot 4 + 2 \cdot 2 \cdot 1 \cdot 0} = \sqrt{11}$$

$$t = \frac{b_1 + 2b_2 - 5}{\text{SE}(b_1 + 2b_2)} \sim t(b_0)$$

$$t_0 = \frac{2 + 2 \cdot 3 - 5}{\sqrt{11}} = \frac{3}{\sqrt{11}}$$

$$RR = \{|t| > 2\}$$

$$\because t_0 = \frac{3}{\sqrt{11}} < 2 \quad \therefore \text{Do not reject } H_0$$

c.

$$H_0: \beta_1 - \beta_2 + \beta_3 = 4$$

$$H_a: \beta_1 - \beta_2 + \beta_3 \neq 4$$

$$\text{SE}(\beta_1 - \beta_2 + \beta_3) = \sqrt{3 + 4 + 3 - 2 \cdot 2 \cdot 1 + 2 \cdot 1 \cdot 0} = 4$$

$$t = \frac{b_1 - b_2 + b_3 - 4}{\text{SE}(b_1 - b_2 + b_3)} \sim t(b_0)$$

$$t_0 = \frac{2 - 3 + (-1) - 4}{4} = \frac{-6}{4} = -\frac{3}{2}$$

$$RR = \{|t| > 2\}$$

$$\because t_0 = -\frac{3}{2} > -2 \quad \therefore \text{Do not reject } H_0$$